QA4SM Automated Validation Report for

SMOS L2 (v700)

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QA4SM version:	3.2.1
Report frequency:	Monthly
Temporal coverage:	91 days (2024-07-01 to 2024-09-30)
Other reports in this series	https://qa4sm.eu/ui/reports
Website:	https://test.qa4sm.eu/
Contact:	support@qa4sm.eu

Note: This report was generated automatically and may contain errors. In case of questions, please contact the platform maintainers via the provided support email address.

1 Report summary

This report contains results from the following validation runs (validation runs are publicly available via the provided URL):

Run	URL	Name	Completed
#1	6eb61199-59b8-4ecc-8e3c-7b1139df4a05	`SMOS vs C3S`	2025-12-01 21:51
#2	e95eeaeb-1d2f-43c4-b019-b7f3b3dbd29e	`SMOS vs ERA5-Land`	2025-12-02 09:25

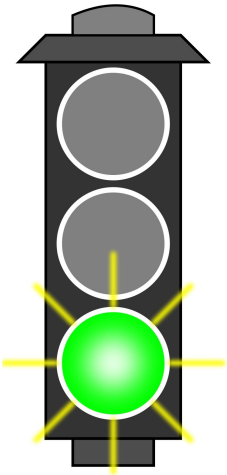

Figure 1: General quality indicator

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2 Datasets

The following chapter contains an overview of the datasets and filters applied in all runs in this report.
The common period for all datasets is from 2024-07-01 to 2024-09-30

2.1 Candidate - SMOS Level 2 (v700) Soil Moisture

The following list contains all data filters applied to SMOS Level 2 before any metrics are computed. For a detailed description and threshold values for each filter, see the validation run online.

- Variable in valid geophysical range.
- Good confidence filtering of RFI.
- Exclude strong topography in scene.
- Exclude presence of frozen ground in scene.
- Exclude presence of ice in scene.
- Exclude presence of dry/wet/mixed snow in scene.
- Exclude open water in scene

2.2 Reference 1 - ERA5-Land (latest) Surface SM

When ERA5-Land is used in a validation run, the following filters are applied before validation metrics are computed:

- Variable in valid geophysical range.
- Soil not frozen

2.3 Reference 2 - C3S SM combined (v202505) Soil Moisture

In all validation runs with C3S SM combined, the following filters are applied to the dataset:

- Variable in valid geophysical range

3 Validation run 1

Fig. 2 shows the extent of validation run 1. All statistics and plots in this chapter are based on data from the shown bounding box.

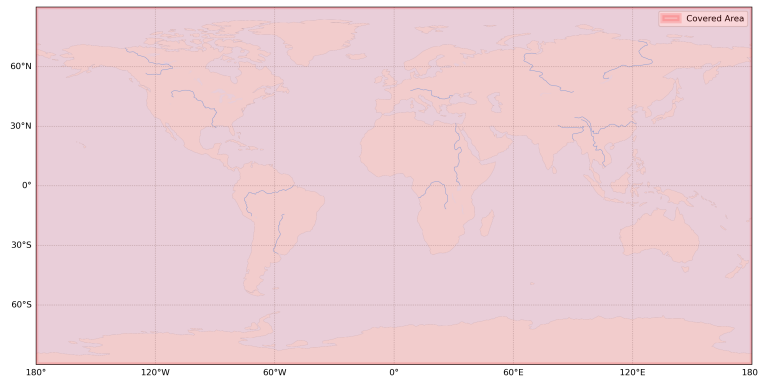

Figure 2: Area covered by all validation runs.

3.1 Validation Settings

The following validation settings were used in the comparison against C3S SM combined. For a full list of used settings, see the validation run results page linked in Table 1.

- Spatial reference: **SMOS Level 2 (v700)**
- Temporal reference: **SMOS Level 2 (v700)**
- Scaling reference: **None**
- Temporal matching window was set to **24 hours**.
- Validation metrics **are not** based on anomalies.
- Triple collocation analysis **was not** performed.

3.2 SMOS L2 Filters

The following data filters were applied to SMOS L2:

1. Variable in valid geophysical range.
2. Good confidence filtering of RFI.
3. Exclude strong topography in scene.
4. Exclude presence of frozen ground in scene.
5. Exclude presence of ice in scene.
6. Exclude presence of dry/wet/mixed snow in scene.
7. Exclude open water in scene
8. Set retrieval quality threshold to 0.05

3.3 C3S SM combined Filters

The following data filters were applied to C3S SM combined:

1. Variable in valid geophysical range

3.4 Results

For 37.00 % of all tested locations (756229) an error was found.

Table 1: Summary statistics of SMOS L2 against C3S SM combined. More metrics can be found online.

Metric	#Obs.	Pearson's R	Bias	ubRMSD
Mean	23.94	0.55	-0.0629	0.0464
Median	0.0	0.53	-0.0664	0.0419
IQ-Range	57.0	0.49	0.0619	0.0325

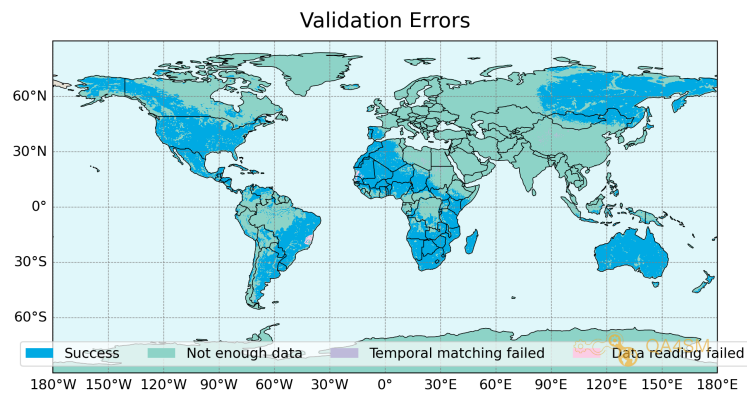

Figure 3: Status Map / Errors raised

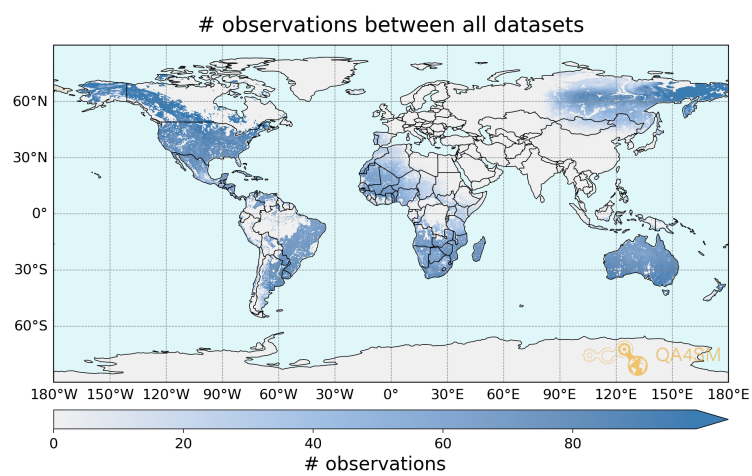

Figure 4: Available Observations

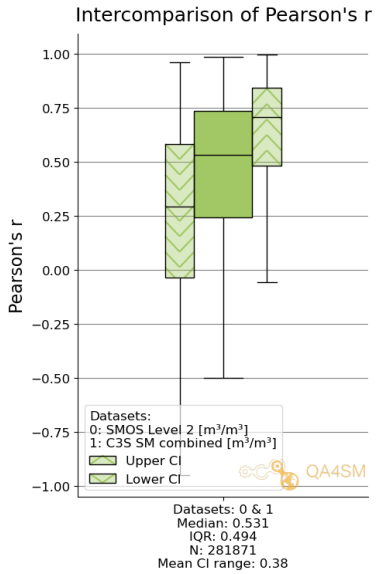

Figure 5: Box plot of Pearson's R

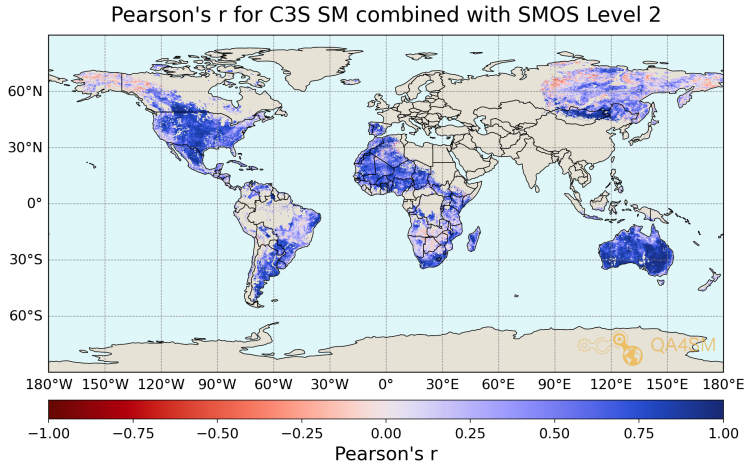

Figure 6: Map of Pearson's R

4 Validation run 2

Fig. 7 shows the extent of validation run 2. All statistics and plots in this chapter are based on data from the shown bounding box.

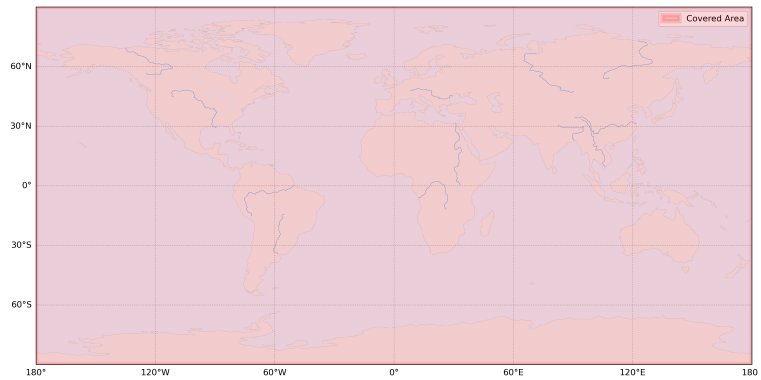

Figure 7: Area covered by all validation runs.

4.1 Validation Settings

The following validation settings were used in the comparison against ERA5-Land. For a full list of used settings, see the validation run results page linked in Table 1.

- Spatial reference: **SMOS Level 2 (v700)**
- Temporal reference: **SMOS Level 2 (v700)**
- Scaling reference: **None**
- Temporal matching window was set to **6 hours**.
- Validation metrics **are not** based on anomalies.
- Triple collocation analysis **was not** performed.

4.2 SMOS L2 Filters

The following data filters were applied to SMOS L2:

1. Variable in valid geophysical range.
2. Good confidence filtering of RFI.
3. Exclude strong topography in scene.
4. Exclude presence of frozen ground in scene.
5. Exclude presence of ice in scene.
6. Exclude presence of dry/wet/mixed snow in scene.
7. Exclude open water in scene
8. Set retrieval quality threshold to 0.05

4.3 ERA5-Land Filters

The following data filters were applied to ERA5-Land:

1. Variable in valid geophysical range.
2. Soil not frozen

4.4 Results

For 44.00 % of all tested locations (756229) an error was found.

Table 2: Summary statistics of SMOS L2 against ERA5-Land. More metrics can be found online.

Metric	#Obs.	Pearson's R	Bias	ubRMSD
Mean	28.26	0.45	-0.1	0.056
Median	0.0	0.38	-0.1	0.0523
IQ-Range	66.0	0.55	0.15	0.0445

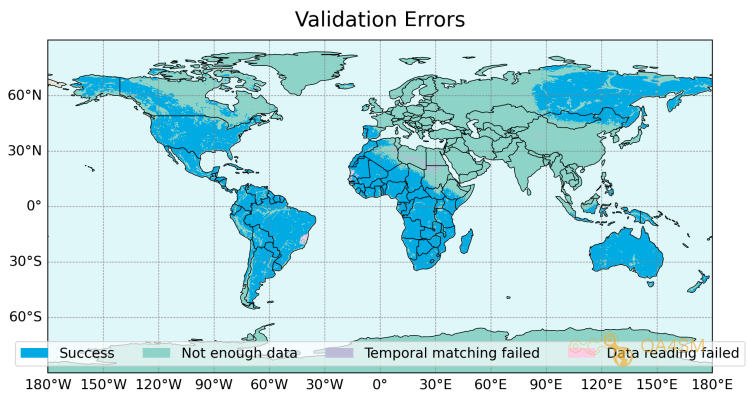

Figure 8: Status Map / Errors raised

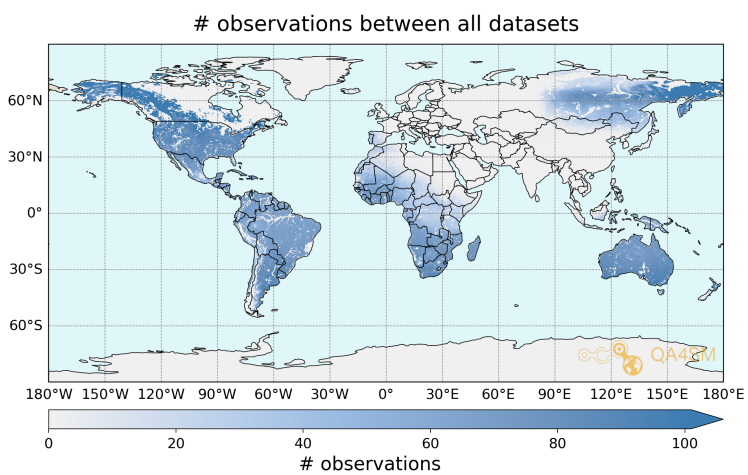

Figure 9: Available Observations

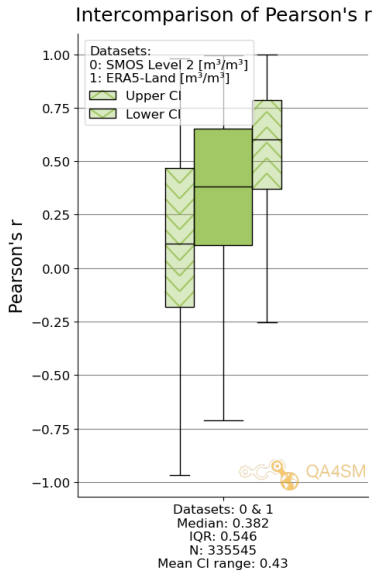

Figure 10: Box plot of Pearson's R

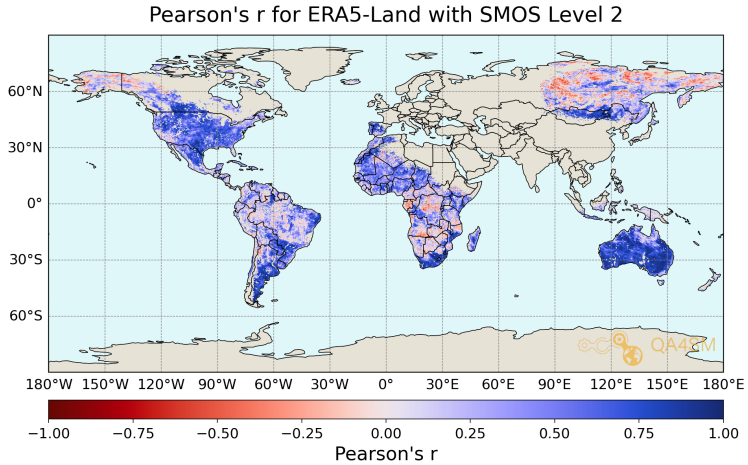

Figure 11: Map of Pearson's R